

Constantinos Veletas

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EDUCATION

University of Colorado at Boulder, B.S. Aerospace Engineering

Aug 2023-May 2027

PROFESSIONAL EXPERIENCE

Astra Core-X, Broomfield, CO May 2026 -

Composite Materials Dev & Propulsion Dev Engineering Intern

- Stood up novel in-house composite component manufacturing effort, from requirements, through design review, to prototyping lab producing components for testing
- Worked to stand up propellant development & production effort via modeling internal ballistics, designing tooling for prop production, mixing & casting prop, static fires & test analysis, and verification of propulsion system through flight testing

Final Frontier Manufacturing, Arvada, CO Jun - Aug 2025

Aerospace Manufacturing Intern

- Fabricated components for rocket engine assemblies, validated and verified quality control during machining & inspection by working between manufacturing engineers, QA lab, and machinists to meet customer tolerances
- Created databases for every piece of instrumentation used by the QA lab to ensure continuous component compliance, up to date instrument calibration, and purchase of new instrumentation for future components

EXTRACURRICULAR EXPERIENCE

CU Sounding Rocket Lab, Boulder, CO Sep 2023 -

Liquids Senior Responsible Engineer

Jun 2025-

- Leading the design of SRL's first additively manufactured rocket engine injector; from initial requirements, through orifice sizing, to MRR
- Manufactured, assembled and tested Diablo engine test stand for use in all current SRL liquid engine tests
- Sized all structural components for multiple ablative liquid engines, confirmed structural component margins; conducting hydrostatic testing to verify margins, validating performance with static fire testing
- Designed ignition system for all current SRL liquid rocket engines, verified ignition modeling through physical testing; Improved design through multiple static fire tests

Solid Rocket Structures, Propellant Team, and Avionics team Technician

Nov 2024-

- Built and tested combination vacuum chamber and vibration table to be used for mixing of all solid propellant used by SRL
- Worked with Composites team to manufacture custom composite kevlar phenolic TPS liners, and carbon fiber Casings on 2-axis winder for Mamba III
- Worked with Avionics team to prep the Avionics bay for vehicle integration and maintain radio communication with Mamba III for successful ascent, descent, and recovery from 90,000 ft

SKILLS

Engineering & Technical: Structural Analysis, Fluid System design, CAD, FEA, Internal Ballistics, GD&T, DFMA, Test Execution,

Software: Matlab | Python | Solidworks | Fusion 360 | Ansys Mechanical | NI LabView | C++

Manufacturing: 3D printing (Polymer & Metal) | Lathe, Mill, & CNC machining | Composites Manufacturing | Propellant Manufacturing